

2014-2015 - Macro 1 - F. PORTIER -

Lecture 1 +

1 Transversality condition (TC)

the TC is a necessary condition for optimality.

For the problem

$$\begin{aligned} & \text{Max} \int_0^T u(k_t, c_t) dt \\ & \text{s.t.} \left\{ \begin{array}{l} k_0 = \bar{k}_0 \\ \dot{k}_t = F(k_t, c_t) \\ k_T \geq 0 \end{array} \right. \end{aligned}$$

The TC is $\lambda_T K_T = 0$

If the terminal condition was

$$K_T > \overline{K_T},$$

then the TC would be $\lambda_T (K_T - \overline{K_T}) = 0$

Note: A terminal condition is given to the problem.

- A transversality condition is necessary for optimality: it is chosen by the problem solver.

2 A two x two example

$$K_i = \begin{pmatrix} k_i \\ s_i \end{pmatrix} \quad G_i = \begin{pmatrix} c_i \\ z_i \end{pmatrix}$$

$$M_i(K_i, G_i) = M_i(k_i, s_i, c_i, z_i)$$

$$K_{i+1} = G(K_i, C_i) \quad i = 0, N$$

$$\Leftrightarrow \begin{cases} K_{i+1} = g(k_i, s_i, c_i, z_i) & \begin{pmatrix} \lambda_{1i} \\ \lambda_{2i} \end{pmatrix} \\ S_{i+1} = h(k_i, s_i, c_i, z_i) & \end{cases}$$

$$\lambda_i = \begin{pmatrix} \lambda_{1i} \\ \lambda_{2i} \end{pmatrix}$$

$$\nabla_{c, G} = \begin{pmatrix} \frac{\partial g}{\partial c} & \frac{\partial g}{\partial z} \\ \frac{\partial h}{\partial c} & \frac{\partial h}{\partial z} \end{pmatrix}$$

$$\nabla_c \mathcal{M} = \begin{pmatrix} \frac{\partial \mathcal{M}}{\partial c} & \frac{\partial \mathcal{M}}{\partial z} \end{pmatrix}$$

$$\nabla_c \mathcal{L} = \begin{pmatrix} \frac{\partial \mathcal{L}}{\partial c} & \frac{\partial \mathcal{L}}{\partial z} \end{pmatrix}$$

The Lagrangian writes

$$\begin{aligned}
 \mathcal{L} = & \sum_{i=0}^N \mathcal{M}_i(k_i, s_i, c_i, z_i) \\
 & + \sum_{i=0}^N \begin{pmatrix} \lambda_{1i} \\ \lambda_{2i} \end{pmatrix}^T \times \begin{pmatrix} g(k_i, s_i, c_i, z_i) - k_{i+1} \\ h(k_i, s_i, c_i, z_i) - s_{i+1} \end{pmatrix} \\
 & + \begin{pmatrix} \overline{\lambda}_1 \\ \overline{\lambda}_2 \end{pmatrix}^T \times \begin{pmatrix} k_{N+1} \\ s_{N+1} \end{pmatrix}
 \end{aligned}$$

Foc are

$$\nabla_{c_i} \mathcal{L} = 0 \Leftrightarrow \nabla_{c_i} \mathcal{M}_i + \lambda_1^T \nabla_{c_i} G = 0$$

$$\Leftrightarrow \begin{cases} \frac{\partial \mathcal{M}_i}{\partial c_i} + \lambda_{1i} \frac{\partial g}{\partial c_i} + \lambda_{2i} \frac{\partial h}{\partial c_i} = 0 \\ \frac{\partial \mathcal{M}_i}{\partial z_i} + \lambda_{1i} \frac{\partial g}{\partial z_i} + \lambda_{2i} \frac{\partial h}{\partial z_i} = 0 \end{cases}$$

and

$$\nabla_{k_i} \mathcal{L} = 0 \Leftrightarrow \nabla_{k_i} \mu_i - \lambda_{i-1}^T + \lambda_i^T \nabla_{k_i} G = 0$$

$$\Leftrightarrow \left\{ \begin{array}{l} \frac{\partial \mu_i}{\partial k_i} - \lambda_{1i-1} + \lambda_{1i} \times \frac{\partial g}{\partial k_i} \\ \quad + \lambda_{2i} \times \frac{\partial h}{\partial k_i} = 0 \\ \\ \frac{\partial \mu_i}{\partial s_i} - \lambda_{2i-1} + \lambda_{1i} \frac{\partial g}{\partial s_i} \\ \quad + \lambda_{2i} \frac{\partial h}{\partial s_i} = 0 \end{array} \right.$$

$$+ \quad \lambda_{1N} = \overline{\lambda}_1$$

$$\lambda_{2N} = \overline{\lambda}_2$$

$$\overline{\lambda}_1 \quad k_{N+1} = 0$$

$$\overline{\lambda}_2 \quad s_{N+1} = 0$$